Fact Schema Design Workflow

Giuseppe Rudi

July 16, 2026

Introduction

This document describes the complete workflow used to design the Fact Schema starting from the reconciled database. The process follows a structured methodology that includes conceptual design, transformation, and logical modeling.

Naming Convention

The attribute tree is built following the data-driven DFM methodology. According to this methodology, the root of the attribute tree formally corresponds to the identifier of the selected fact.

In the reconciled schema, `lap_id` and `result_id` identify source records and support lineage and data-quality checks. They are not exported as analytical fact columns. The physical warehouse facts use `lap_performance_key` and `session_result_key` as their primary keys.

However, for graphical readability, the root of the attribute tree and the DFM fact schema may be labelled with a conceptual name such as *Lap Performance*. This label must be interpreted as a readable representation of the fact identifier, while the formal construction of the attribute tree remains based on the identifier of the selected fact.

The DFM fact schema is a conceptual multidimensional model. Its goal is to describe the fact, the dimensions, the hierarchies, and the measures from an analytical point of view. For this reason, it may use readable business-oriented names.

The star schema, instead, is the logical translation of the DFM into relational tables. For this reason, the star schema must use the implementation-oriented naming convention adopted in the database, including `snake_case` table names, attributes, primary keys, foreign keys, and measures.

Starting Point: Reconciled Database

The reconciled database remains the correct operational starting point of the project. It already organizes the source data into a coherent relational structure and should not be considered wrong just because some tables are not directly transferable into a single fact schema.

The process starts from the reconciled database, which contains normalized and integrated data.

In our case, the reconciled schema includes entities such as:

- Lap
- Driver
- Team
- Session
- Grand Prix

- Weather
- Track Status

The reconciled schema represents the operational structure and is not optimized for analytical queries.

In particular:

- the **Lap** relation is the reconciled representation of lap-level events;
- the **Result** relation is the reconciled representation of final results at session level;

Therefore, the reconciled database must be interpreted as the *source structure* for conceptual design, not as a structure that must be copied one-to-one into the fact schema.

The reconciled schema remains normalized and contains `season`, `circuit`, `grand_prix`, `session`, `driver`, `team`, `result`, `lap`, `weather`, and `track_status`. The denormalization described later applies only to the physical warehouse.

The final physical warehouse contains exactly the following tables:

- `fact_lap_performance` and `fact_session_result`;
- `dim_driver`, `dim_team`, and `dim_grand_prix`;
- `dim_tyre_context`, `dim_lap_weather_context`, and `dim_lap_data_quality`;
- `dim_result_outcome`, `dim_result_weather_context`, and `dim_result_data_quality`.

No physical `dim_circuit`, `dim_season`, `dim_session_type`, `dim_lap_type`, `dim_track_status_context`, or generic `dim_weather_context` is created.

A fact must represent a meaningful analytical event, and its grain must be declared before defining dimensions, measures, and hierarchies.

Once the grain is fixed, only information coherent with that grain can be attached directly to the fact.

Fact 1: Lap Performance

This is the most natural fact for the core analytical goal of the project, namely the comparison of performance at lap level, enriched by sector information.

Grain

The grain of this fact is:

> One lap performed by one driver in one specific session, belonging to one specific Grand Prix and one specific Formula 1 season.

Operationally, this fact is rooted in the reconciled `Lap` table.

Fact 2: Session Result

A second fact is necessary because not all relevant analyses of the project are lap-level analyses.

Some questions concern the **final behaviour of a driver, team, or car in a session or along a season**, not the performance of a single lap.

Examples include:

- comparison of qualifying outcomes;
- comparison of race outcomes;

- season-level summary of driver or team results;

The second fact should be based on **Result**.

Grain

The grain of this fact is:

> One driver in one specific session, belonging to one specific Grand Prix and one specific Formula 1 season.

Operationally, this fact is rooted in the reconciled `result` table.

Expected Results

Attribute Trees and Editing

The next conceptual consequence is straightforward:

> Two facts imply two attribute trees, two editing processes, and two fact schemas.

Therefore, the project should produce:

- a DFM for **Lap Performance**;
- a DFM for **Result**.

These two schemas should not be considered isolated. They should be connected through shared dimensions such as Driver, Team, Session, Grand Prix, Circuit, Season. This will later support integrated analyses across the two facts.

Once the two facts are defined, it becomes possible to support both separate and combined analyses:

- **Lap Performance** for fine-grained comparison of laps and sectors;
- **Session Result** for broader outcome-oriented analysis of drivers and teams;
- shared analyses through common dimensions when the two facts must be read together.

This means that the final data warehouse should be designed as a multidimensional model with **multiple facts and shared dimensions**, not as a single monolithic fact table trying to answer every type of analytical question.

Weather and Track Status

The *Weather* and *Track Status* relations are not represented as direct branches in the formal attribute trees of the *Lap Performance* fact and the *Session Result* fact.

This choice is due to the absence of a valid to-one branch. In the reconciled database, both relations are time-dependent at session level: one session can contain multiple weather observations and multiple track-status changes. Therefore, a fact instance does not directly determine one unique weather or track-status record.

Nevertheless, these relations remain analytically important. Their information can be used during the ETL process to derive new attributes coherent with the grain of each fact. In particular, weather and track-status data can produce lap-level derived attributes for *Lap Performance* and session-level derived attributes for *Session Result*.

For this reason, the following sections describe these two source relations and the attributes that may be used to build derived analytical attributes.

Weather

Attribute	Meaning
AirTemp	Air temperature measured during the session.
Humidity	Relative humidity measured during the session.
Pressure	Atmospheric pressure measured during the session.
Rainfall	Boolean flag indicating whether rainfall was detected.
TrackTemp	Track surface temperature measured during the session.
WindDirection	Wind direction measured during the session.
WindSpeed	Wind speed measured during the session.

Track Status

Attribute	Meaning
Status	Numeric code associated with the track status event.
Message	Textual description of the track status event.

Main categories of the `Message` attribute

Message	Meaning
AllClear	The track is in normal condition and no special restriction is active.
Yellow	Yellow flag condition on track, usually caused by an incident, stopped car, or local danger.
SCDeployed	The Safety Car has been deployed.
Red	Red flag condition. The session is stopped or suspended.
VSCDeployed	The Virtual Safety Car has been deployed.
VSCEnding	The Virtual Safety Car phase is ending.

WorkFlow Lap Performance Fact

Relations retained

The following relations are retained in the attribute tree because they are structurally coherent with the grain of the fact and support the intended analytical dimensions and hierarchies.

Relation	Motivation
Lap	It is the source relation of the chosen fact
Driver	Each lap is performed by exactly one driver.
Team	Each lap is associated with exactly one team.
Session	Each lap belongs to exactly one session.
Grand Prix	Each session belongs to exactly one Grand Prix.
Circuit	Each Grand Prix is associated with exactly one circuit.
Season	Each Grand Prix belongs to exactly one Formula 1 season.

Relations excluded

The following relations are part of the reconciled schema, but they are not included in the formal attribute tree of *Lap Performance*. The reason is that they do not define valid branches under the attribute-tree construction rules, since the parent node does not determine a unique child instance.

Relation	Motivation
Weather	A session is associated with multiple weather observations collected at different times.
TrackStatus	A session is associated with multiple track-status changes over time.
Result	A session contains one result for each driver

Derived lap-level attributes from Weather and TrackStatus

Although the relations *Weather* and *TrackStatus* are excluded from the formal attribute tree of *Lap Performance*, the information they contain remains analytically important for the project.

In particular, weather conditions and track status may strongly affect lap performance. For this reason, they are still relevant when the goal is to classify, compare, or filter laps according to the external conditions under which they were performed.

However, in the reconciled database these two relations are modeled as *session-level time-dependent data*. This means that a session is associated with multiple weather observations and multiple track-status changes, each identified by a different time instant.

For this reason, the adopted solution is not to connect *Weather* and *TrackStatus* directly to *Session* inside the attribute tree. Instead, the chosen solution is to derive a set of new lap-level descriptive attributes and attach them directly to *Lap Performance*.

Operationally, this transformation is carried out during the ETL process. The ETL uses `lap_start_time_ms` to associate each lap with the most recent weather observation and track-status record in the same session. A backward temporal association is attempted first; when no preceding observation exists, the nearest record in that same session is used as fallback. Session boundaries are never crossed. In this way, session-level temporal data are transformed into lap-level attributes.

Derived attributes are:

- *AirTempClass*
- *TrackTempClass*
- *RainFlag*
- *WindSpeedClass*
- *TrackStatusCategory*

Pruning and Grafting Operations

Lap Table

We indicate in this table only the descriptive attributes not full of them. In particular the composte primary key attribute are not reported.

Attribute	Meaning
Time_ms	Elapsed session time, in milliseconds, at which the lap record
Driver	Driver abbreviation (for example, `GAS` for Pierre Gasly).
DriverNumber	Official racing number of the driver for that session.
LapTime_ms	Total duration of the completed lap, expressed in milliseconds.
LapNumber	Sequential number of the lap completed by the driver.
Stint	Tyre stint number to which the lap belongs. A new stint usually starts after a pit stop.
PitOutTime_ms	Session elapsed time at which the driver exited the pit lane before or during the lap. Usually null for standard laps.

Attribute	**Meaning**
PitInTime_ms	Session elapsed time at which the driver entered the pit lane at the end of or during the lap. Usually null for standard laps.
Sector1Time_ms	Time spent in Sector 1 of the lap
Sector2Time_ms	Time spent in Sector 2 of the lap
Sector3Time_ms	Time spent in Sector 3 of the lap
Sector1SessionTime_ms	Session elapsed time at which the driver completed Sector 1 of the lap.
Sector2SessionTime_ms	Session elapsed time at which the driver completed Sector 2 of the lap.
Sector3SessionTime_ms	Session elapsed time at which the driver completed Sector 3 of the lap and therefore completed the whole lap.
SpeedI1	Speed measured at the first intermediate speed trap of the circuit.
SpeedI2	Speed measured at the second intermediate speed trap of the circuit.
SpeedFL	Speed measured at the finish line.
SpeedST	Speed measured at the main speed trap, usually the highest-speed checkpoint on the track.
IsPersonalBest	this lap is the driver's personal best lap within that session up to that point.
Compound	Tyre compound used during the lap, such as `SOFT`, `MEDIUM`, or `HARD`.
TyreLife	Number of laps already completed on that tyre set when this lap was run.
FreshTyre	Boolean flag indicating whether the tyre set was new when fitted to the car.
Team	Team name
LapStartTime_ms	Session elapsed time at which the lap started.
LapStartDate	Absolute timestamp of the lap start, when available.
TrackStatus	Track status code associated with the lap context, for example normal conditions, yellow flag, safety car, or other control conditions.
Position	Driver's race position associated with that lap
Deleted	Boolean flag indicating whether the lap was deleted or invalidated in the source data.
DeletedReason	Textual explanation, when available, describing why the lap was deleted or invalidated.
FastF1Generated	Boolean flag indicating that the value or record was generated or reconstructed internally by FastF1 rather than directly available from the source feed.
IsAccurate	Boolean flag indicating whether FastF1 considers the lap timing information reliable and accurate enough for analysis.

Pruning Operations :

Attribute	**Motivation**
PitOutTime_ms	it does not describe the competitive performance of a lap itself.
PitInTime_ms	it does not measure the intrinsic quality of a lap.
Sector1SessionTime_ms	removed because it is a technical timing reference to the session timeline and not essential for the selected business analyses.
Sector2SessionTime_ms	removed because it is a technical timing reference to the session timeline and not essential for the selected business analyses.

Attribute	Motivation
Sector3SessionTime_ms	removed because it is a technical timing reference to the session timeline and not essential for the selected business analyses.
LapStartDate	not usefull to know the date because this information it also reported in the session table
Deleted	removed because it is treated as a data quality control flag
TrackStatus	removed because as it is it does not provide precise and categorical information
DeletedReason	removed because it is mainly useful for cleaning and validation tasks
FastF1Generated	removed because it is a technical flag
IsAccurate	removed because it is relevant for data quality assessment.
DriverNumber	removed because the analytical model already identifies the driver through DriverId and the related Driver dimension, making this attribute redundant.
Team	removed because the team is already represented structurally through TeamId and the Team dimension, so the textual field would introduce redundancy.
Time_ms	Removed because the analysis focuses on lap-order-based performance rather than on absolute session timing
LapStartTime_ms	Removed because it represents a technical session timestamp that does not add analytical value in a lap-based model.
FreshTyre	Removed from the dimensional model in its raw boolean form because its meaning is potentially ambiguous. I
TyreLife	Removed from the dimensional model in its raw numerical form because it would introduce too many distinct values in the tyre-related dimension.
Stint	Removed because the analysis does not focus on the progressive stint number itself.
IsPersonalBest	Removed because it is not central to the analytical goal of the project and its semantic interpretation may be ambiguous.

Although `FreshTyre` and `TyreLife` are pruned from the dimensional model in their original form, their analytical meaning is not discarded.

In the dedicated section, these attributes are used to derive two categorical attributes, namely *StartingTyreSetStatus* and *TyreLifeClass*, which are more suitable for the final *Tyre Context* dimension.

Derived attribute for lap filtering. Even though *PitOutTime_ms* and *PitInTime_ms* are removed from the formal attribute tree, their information is not lost. Instead, they can be used during the ETL process to derive a new categorical attribute that indicates whether a lap is a normal lap, an out-lap, or an in-lap. This derived attribute is useful to exclude pit-affected laps from analyses focused on fast and clean racing laps, thus improving the quality and interpretability of lap-performance comparisons.

Driver Table

We indicate in this table only the descriptive attributes. In particular, the identifier attribute is not reported.

Attribute	Meaning
PermanentNumber	Official permanent racing number assigned to the driver in Formula 1.
Abbreviation	Standard three-letter abbreviation used to identify the driver in Formula 1 timing and result systems.

Attribute	Meaning
DriverUrl	URL of the reference page associated with the driver in the source data.
FirstName	Driver's first name.
LastName	Driver's last name.
DateOfBirth	Driver's date of birth.
Nationality	Nationality associated with the driver.

Retained Attributes Motivation

- **Abbreviation** was retained because, although it is not a core analytical attribute, it is useful as a compact descriptive label in plots, tables, and dashboards.

Pruning Operations

Attribute	Motivation
PermanentNumber	Removed because it mainly acts as a display-oriented sporting identifier
DateOfBirth	it does not provide a directly meaningful analytical level
DriverUrl	Removed because it is an external reference link and has no analytical value

Potential Derived Attributes

Although **DateOfBirth** is removed from the final attribute tree, its information is not necessarily lost. It may be used in a transformation step to derive more analytically meaningful attributes, such as age-based driver categories.

For example, a derived attribute such as *DriverAgeClass* could classify drivers into groups like younger, intermediate, and senior drivers. This type of categorization could be useful to investigate whether drivers of different age groups adapted differently to the 2022 regulatory change.

A second potentially interesting classification would be based on Formula 1 experience rather than biological age. However, this attribute cannot be derived from the current *Driver* table alone, because the table does not include information about the number of seasons completed by each driver in Formula 1. Such an attribute would require additional domain knowledge or an extra relation in the reconciled database.

Team Table

We indicate in this table only the descriptive attributes. In particular, the identifier attribute is not reported.

Attribute	Meaning
TeamUrl	External web reference associated with the team
TeamName	Official name of the Formula 1 team.
Nationality	Nationality associated with the team.

Pruning Operations

Attribute	Motivation
TeamUrl	Removed because it is an external reference link

Season Table

We indicate in this table only the descriptive attributes. In particular, the identifier attribute is not reported.

Attribute	Meaning
SeasonStartDate	Official starting date of the Formula 1 season.
SeasonEndDate	Official ending date of the Formula 1 season.
NumberOfEvents	Total number of championship events included in the season.
ChampionDriver	Driver who won the Drivers' Championship in that season.
ChampionTeam	Team that won the Constructors' Championship in that season.

Retained Attributes Motivation

- Only **SeasonYear** is retained as a level of the Grand Prix hierarchy. The championship descriptors are not required by either analytical fact.

Pruning Operations

Attribute	Motivation
SeasonStartDate	The exact calendar start date does not add relevant value to the intended analyses.
SeasonEndDate	The exact calendar end date does not add relevant value to the intended analyses.
NumberOfEvents	The analysis does not focus on the overall size of the season.
ChampionDriver	Removed because it describes the final championship outcome rather than the grain of either fact.
ChampionTeam	Removed because it describes the final championship outcome rather than the grain of either fact.

IMPORTANT NOTE : The attributes `champion_driver` and `champion_team` were pruned from the attribute tree of the *Lap Performance* fact. This choice was made because these attributes do not provide information about the performance of a specific lap.

The *Lap Performance* fact focuses on lap-level analysis, such as lap time, sector times, tyre compound, session type, Grand Prix, circuit characteristics, driver, and team. From this point of view, knowing the final champion of the season does not directly help to explain the behaviour or performance of a single lap.

The same pruning choice applies to the *Session Result* fact. Seasonal analysis is still supported through `season_year`, which is retained in the Grand Prix hierarchy and physically stored in `dim_grand_prix`.

Grand Prix Table

We indicate in this table only the descriptive attributes. In particular, the identifier attributes and foreign keys are not reported.

Attribute	Meaning
RoundNumber	numerical order from 1 to the total number of Grand Prix in a season
EventName	Short and readable name of the Formula 1 Grand Prix.
OfficialEventName	Full official commercial name of the Formula 1 Grand Prix.
Country	Country in which the Grand Prix takes place.
Location	Geographic location associated with the Grand Prix.
EventDate	Official date of the Grand Prix event.

Attribute	Meaning
EventFormat	Format of the race weekend, for example conventional or sprint.
F1ApiSupport	Boolean flag indicating whether the official F1 API provides session timing support for that event.

Retained Attributes Motivation

- **EventName** was retained because it provides the most useful and readable business name for identifying each Grand Prix in the analytical model.
- **EventDate** was retained because it provides a direct temporal reference for the Grand Prix
- **EventFormat** was retained because it may be useful to distinguish conventional weekends from sprint weekends and therefore support homogeneous analytical filtering.

Pruning Operations

Attribute	Motivation
OfficialEventName	Removed because it is too verbose and commercial in style, while EventName already provides a clearer identification
Country	Removed because this geographic information is already represented in the Circuit table
Location	Removed because this geographic information is already represented in the Circuit table
F1ApiSupport	Removed because it is a technical source-level flag.

Circuit Table

We indicate in this table only the descriptive attributes. In particular, the identifier attribute is not reported.

Attribute	Meaning
CircuitName	Official name of the Formula 1 circuit.
Country	Country in which the circuit is located.
Location	Geographic location of the circuit.
SeasonsPresent	List of seasons in which the circuit appears in the project scope.
GlobalCircuitCategory	General technical category assigned to the circuit, such as Street, PowerSensitive, AeroSensitive, or Mixed.
Sector1Category	Technical category assigned to Sector 1 of the circuit.
Sector2Category	Technical category assigned to Sector 2 of the circuit.
Sector3Category	Technical category assigned to Sector 3 of the circuit.

Pruning Operations

Attribute	Motivation
SeasonsPresent	Removed because the presence of a circuit across seasons can already be inferred from the relationships between Season and Grand Prix

Session Table

We indicate in this table only the descriptive attributes. In particular, identifier attributes are generally not reported. However, *SessionType* is included here because, although it is part of the composite identifier, it also conveys a relevant semantic meaning for the analysis by distinguishing the type of session.

Attribute	Meaning
SessionType	Type of Formula 1 session, such as `Q` for Qualifying or `R` for Race.
SessionName	Full textual name of the session, such as `Qualifying` or `Race`.
SessionDate	Calendar date on which the session took place.

Pruning Operations

Attribute	Motivation
SessionName	Removed because it is semantically redundant with *SessionType*
SessionDate	Removed because the exact calendar date of the session is not central already available through the related Grand Prix.

Grafting Operation

After the pruning operations, a grafting operation was applied to the `Session_Id` node.

In the *Lap Performance* workflow, the root of the attribute tree is `Lap_Id`.

Before grafting, the fact was connected to the session branch through `Session_Id`. The `Session_Id` node represented the session entity and connected the lap-level fact to the higher-level event hierarchy.

The initial structure was:

`Lap_Id` → `Session_Id` → `GrandPrix_Id`

After the pruning operations, the only analytically relevant attribute retained from the session node was `SessionType`. Since `Session_Id` is mainly a technical identifier and does not represent an analytical level of interest by itself, it was removed through grafting.

After grafting, the useful descendants of `Session_Id` are connected directly to `Lap_Id`. Therefore, `SessionType` becomes a direct attribute of `Lap_Id`, and `GrandPrix_Id` becomes a direct child of `Lap_Id`.

The resulting structure is:

`Lap_Id` → `SessionType`

and

`Lap_Id` → `GrandPrix_Id`.

Derived Attributes

After the pruning and grafting operations, some additional attributes are introduced as derived attributes.

These attributes are not directly taken as branches from the reconciled schema. Instead, they are computed during the ETL process from source relations or from pruned technical attributes.

The reason is that some useful analytical information is available in the reconciled database, but it is not directly represented at the same grain of the *Lap Performance* fact.

Therefore, every derived attribute attached to this fact must also be transformed to the lap-level grain.

The *Weather* relation is not included as a direct branch of the formal attribute tree. The reason is that weather data are recorded several times during a session. Therefore, one session is associated with multiple weather observations.

This means that the relationship:

`Session_Id` → `Weather`

does not determine a single weather record. For this reason, the raw *Weather* relation cannot be directly attached to the *Lap Performance* attribute tree.

However, weather information is analytically relevant because environmental conditions can influence lap performance, tyre behaviour, grip level, and track evolution.

For this reason, the ETL process derives lap-level weather attributes by associating each lap with the weather observation valid during that lap interval.

Conceptually, the ETL uses the lap time information, such as the lap start time and the lap end time, to find the corresponding weather observation within the same session. After this transformation, the weather information becomes coherent with the grain of the *Lap Performance* fact.

The original weather attributes *AirTemp*, *TrackTemp*, and *WindSpeed* are continuous numerical values. Although they could be stored as contextual numerical measures, in the current Data Warehouse design they are transformed into categorical dimensional attributes.

This choice is motivated by the analytical goal of the project. The analysis is not based on exact temperature or wind-speed values, but on comparisons between laps performed under similar environmental conditions. Therefore, fixed ranges are defined during the ETL process and used to derive the attributes *AirTempClass*, *TrackTempClass*, and *WindSpeedClass*.

The original numerical values are preserved in the reconciled database, while the Data Warehouse stores their categorical representation. This avoids unnecessary numerical detail in the fact table and makes OLAP queries simpler and more consistent.

Derived Attribute	**Source Attribute**	**Analytical Meaning**
AirTempClass	`weather.air_temp`	Air temperature associated with the lap.
TrackTempClass	`weather.track_temp`	Track surface temperature associated with the lap.
RainFlag	`weather.rainfall`	Boolean flag indicating whether the lap was performed under rainy or wet conditions.
WindSpeedClass	`weather.wind_speed`	Wind speed associated with the lap.

The derived attribute *AirTempClass*, *TrackTempClass*, *WindSpeedClass* are attached directly to `Lap_Id`

Possible values are:

Derived Attribute	**Possible Values**	**Meaning**
AirTempClass	*Low*, *Medium*, *High*, *Unknown*	Classifies the air temperature associated with the lap into fixed ranges. It is useful to compare laps performed under similar ambient temperature conditions.
TrackTempClass	*Low*, *Medium*, *High*, *Unknown*	Classifies the track temperature associated with the lap. It is useful because track temperature can influence tyre behaviour and grip.
WindSpeedClass	*Low*, *Medium*, *High*, *Unknown*	Classifies the wind speed associated with the lap. It helps distinguish laps performed in more stable wind conditions from laps performed in stronger wind conditions.

Derived Attribute	Class	Initial Rule
AirTempClass	*Low*	$AirTemp < 20°C$
AirTempClass	*Medium*	$20°C \leq AirTemp < 28°C$
AirTempClass	*High*	$AirTemp \geq 28°C$
AirTempClass	*Unknown*	*AirTemp* is missing or unreliable
TrackTempClass	*Low*	$TrackTemp < 30°C$
TrackTempClass	*Medium*	$30°C \leq TrackTemp < 42°C$
TrackTempClass	*High*	$TrackTemp \geq 42°C$
TrackTempClass	*Unknown*	*TrackTemp* is missing or unreliable
WindSpeedClass	*Low*	$WindSpeed < 2$
WindSpeedClass	*Medium*	$2 \leq WindSpeed < 5$
WindSpeedClass	*High*	$WindSpeed \geq 5$
WindSpeedClass	*Unknown*	*WindSpeed* is missing or unreliable

These attributes are attached directly to `Lap_Id` in the attribute tree, because after the ETL process they describe the specific conditions under which each lap was performed.

The *TrackStatus* relation is also not included as a direct branch of the formal attribute tree.

The reason is similar to the weather case. Track status data are time-dependent: during the same session, the track can move through different states, such as normal condition, yellow flag, safety car, virtual safety car, or red flag.

Therefore, a session does not determine a single track-status record.

However, track status is important for lap-performance analysis because a lap performed under yellow flag, safety car, or red flag conditions is not directly comparable with a lap performed under normal racing conditions.

For this reason, the raw track-status information is transformed into lap-level categorical attributes during the ETL process.

Derived Attribute	Source Attribute	Analytical Meaning
TrackStatusCategory	`track_status.message`	categorical description of the track-status message, such as all-clear, yellow flag, safety car, virtual safety car, or red flag.

These attributes are attached directly to `Lap_Id`, because after the ETL process each lap is associated with the track condition valid during that lap.

Some technical attributes of the *Lap* relation were pruned from the formal attribute tree because they are not useful as analytical dimensions or measures.

This is the case of `PitOutTime_ms` and `PitInTime_ms`. These attributes do not directly describe the competitive quality of a lap, but they are still useful during the ETL process because they allow the identification of pit-affected laps.

For this reason, their information is used to derive a new lap-level categorical attribute.

Derived Attribute	Source Attribute	Analytical Meaning
LapType	`pit_out_time_ms`, `pit_in_time_ms`	Categorical attribute used to distinguish normal laps, out-laps, in-laps, and pit-affected laps. It is useful to filter the dataset when the analysis focuses on clean performance laps.

The derived attribute *LapType* is attached directly to `Lap_Id`, because it describes the nature of the specific lap.

Possible values are:

- *NormalLap*
- *OutLap*
- *InLap*
- *PitAffectedLap*

This derived attribute is particularly useful because out-laps and in-laps usually do not represent pure performance laps. Therefore, they should often be excluded from analyses focused on lap-time comparison.

The original lap data contain two tyre-related attributes that are not directly used in their raw form in the final dimensional model: *TyreLife* and *FreshTyre*.

The attribute *TyreLife* is a numerical value representing the number of laps already completed on the tyre set, including laps completed in previous sessions when the same tyre set was already used. Although this attribute is useful to understand tyre usage, it is not suitable as a dimensional attribute in its original numerical form. Using the exact value of *TyreLife* would generate too many possible combinations in the tyre context dimension and would not reflect the intended analytical usage of the model.

For this reason, the raw attribute *TyreLife* is pruned from the dimensional model and replaced by the derived categorical attribute *TyreLifeClass*. This attribute groups tyre usage into the following closed classes:

- *Low*: $0 \leq \texttt{tyre\_life} \leq 5$;
- *Medium*: $5 < \texttt{tyre\_life} \leq 15$;
- *High*: $\texttt{tyre\_life} > 15$;
- *Unknown*: value missing, negative, or unreliable.

The attribute *FreshTyre* is also not used directly in the final dimensional model. In the source data, *FreshTyre* indicates whether the tyre set had *TyreLife* equal to zero at the beginning of the stint. Therefore, it does not mean that the tyre is necessarily new in the current lap, but rather that the tyre set was new when it was mounted at the start of the stint.

For this reason, the raw attribute *FreshTyre* is pruned and replaced by the derived categorical attribute *StartingTyreSetStatus*. This attribute provides a clearer semantic interpretation and distinguishes between tyre sets that were new or already used at stint start.

- *NewSet*: the tyre set was new at the beginning of the stint, i.e., `FreshTyre = True`;
- *UsedSet*: the tyre set was already used at the beginning of the stint, i.e., `FreshTyre = False`;
- *Unknown*: the initial status of the tyre set is not available or the value of `FreshTyre` is missing.

The resulting tyre-related derived attributes are therefore:

Derived Attribute	Source Attribute	Analytical Meaning
StartingTyreSetStatus	`FreshTyre`	Categorical attribute used to clarify whether the tyre set was new or already used at the beginning of the stint. It replaces the raw boolean attribute `FreshTyre` with a more explicit semantic interpretation, such as *NewSet*, *UsedSet*, or *Unknown*.
TyreLifeClass	`TyreLife`	Categorical attribute used to group the numerical tyre life into usage ranges.

These derived attributes are then included in the *Tyre Context* dimension together with the original categorical attribute *Compound*.

The attribute `DateOfBirth` was pruned from the formal attribute tree in *driver* table because the exact birth date is not directly useful as an analytical level for lap-performance analysis.

However, it can be used to derive a more meaningful categorical attribute.

Derived Attribute	Source Attribute	Analytical Meaning
DriverAgeClass	`driver.date_of_birth`	Categorical attribute that groups drivers into age classes. It may support optional analyses about whether drivers of different age groups adapted differently to the 2022 regulatory change.

Driver age is computed with the fixed reference date 1 January 2022 as $\lfloor(\texttt{2022-01-01} - \texttt{date\_of\_birth})/365.2425\rfloor$. Possible values are:

- *Young*: age below 25;
- *Intermediate*: age from 25 to 29;
- *Senior*: age at least 30;
- *Unknown*: birth date missing or unreliable.

It is not central to the main business question, which is mainly focused on team and car performance. However, it can be retained as a possible secondary analytical attribute if driver-level comparisons become relevant.

DriverAgeClass should be attached to the `Driver_Id` branch, because it describes the driver rather than the lap itself.

Derived Quality Attributes from Data Cleaning

Some additional lap-level attributes are introduced after the Data Cleaning phase.

These attributes are not part of the original reconciled schema and they are not included in the initial General Attribute Tree. They are derived from the outputs of the focused Missing Values analysis, the Outlier Detection script, and the Data Cleaning workflow.

Their purpose is not to describe a Formula 1 sporting property directly, but to describe the analytical usability of each lap.

For this reason, they are considered *derived quality attributes*.

The Data Cleaning script creates one boolean flag for each relevant analytical information area. A value equal to `true` means that the corresponding information area is missing, incomplete, or unreliable for that lap. A value equal to `false` means that no issue was detected for that information area.

These attributes are attached to `Lap_Id`, because each flag describes the quality condition of one specific lap row.

Derived Quality Attribute	**Analytical Meaning**
HasSectorInformationIssue	Indicates that one or more sector-time attributes are missing, invalid, inconsistent, or detected as unreliable. The lap should not be used for complete sector analysis.
HasSpeedInformationIssue	Indicates that one or more speed attributes are missing, invalid, or detected as unreliable. The lap should not be used for speed-based analysis.
HasTyreInformationIssue	Indicates that tyre-related information, such as compound, stint, or tyre life, is missing or unreliable. The lap should be used carefully in tyre-based comparisons.
HasWeatherInformationIssue	Indicates that lap-level weather context is missing or unreliable. The lap should not be used for weather-based comparison without caution.
HasTrackStatusInformationIssue	Indicates that track-status information is missing or not interpretable. This affects clean-lap filtering and track-condition analysis.
HasPitInformationIssue	Indicates that pit-related information is incomplete. This affects the interpretation of normal laps, in-laps, out-laps, and pit-affected laps.

The following table summarizes the derived attributes considered for the *Lap Performance* fact.

Derived Attribute	**Derived From**	**Attached To**
AirTempClass	*Weather*	`Lap_Id`
TrackTempClass	*Weather*	`Lap_Id`
RainFlag	*Weather*	`Lap_Id`
WindSpeedClass	*Weather*	`Lap_Id`
TrackStatusCategory	*TrackStatus*	`Lap_Id`
StartingTyreSetStatus	`fresh_tyre`	`Lap_Id`
TyreLifeClass	`tyre_life`	`Lap_Id`
LapType	`pit_out_time_ms`, `pit_in_time_ms`	`Lap_Id`
HasSectorInformationIssue	Data Cleaning	`Lap_Id`
HasSpeedInformationIssue	Data Cleaning	`Lap_Id`
HasTyreInformationIssue	Data Cleaning	`Lap_Id`
HasWeatherInformationIssue	Data Cleaning	`Lap_Id`
HasTrackStatusInformationIssue	Data Cleaning	`Lap_Id`
HasPitInformationIssue	Data Cleaning	`Lap_Id`
DriverAgeClass	`driver.date_of_birth`	`Driver_Id`

Dimension Definition

After the pruning and grafting operations, and after the introduction of the derived attributes, the final edited attribute tree is used to define the dimensions of the *Lap Performance* fact.

In the Dimensional Fact Model, dimensions are the analytical coordinates used to study the fact from different points of view.

The following table summarizes the main dimensions selected for the *Lap Performance* fact.

Dimension	**Attributes**	**Motivation**
Driver	`Driver_Id`, *FirstName*, *LastName*, *Abbreviation*, *Nationality*, *DriverAgeClass*	This dimension is used to compare lap performance among drivers.
Team	`Team_Id`, *TeamName*, *Nationality*	The main analytical goal is to compare the competitive performance of teams
Grand Prix	`GrandPrix_Id`, *EventName*, *RoundNumber*, *EventDate*, *EventFormat*, *SeasonYear*, *CircuitId*, *CircuitName*, *Country*, *Location*, *GlobalCircuitCategory*, *Sector1Category*, *Sector2Category*, *Sector3Category*	This dimension allows the analysis of lap performance by grand prix, season, calendar order, and weekend format.
Circuit hierarchy level	`Circuit_Id`, *CircuitName*, *Country*, *Location*, *GlobalCircuitCategory*, *Sector1Category*, *Sector2Category*, *Sector3Category*	This is a conceptual hierarchy in the DFM. The attributes are physically stored inside `dim_grand_prix`; no `dim_circuit` table is created.
Session Type (DD)	*SessionType*	This dimension distinguishes qualifying sessions from race sessions.
Tyre Context	*Compound*, *StartingTyreSetStatus*, *TyreLifeClass*	This dimension describes the tyre condition of the lap.
Track Status (DD)	*TrackStatusCategory*	This dimension is mainly used for filtering the track condition.
Lap Weather Context	*RainFlag*, *AirTempClass*, *TrackTempClass*, *WindSpeedClass*	Physical dimension `dim_lap_weather_context`, referenced through `lap_weather_context_key`.
Lap Data Quality	*HasSectorInformationIssue*, *HasSpeedInformationIssue*, *HasTyreInformationIssue*, *HasWeatherInformationIssue*, *HasTrackStatusInformationIssue*, *HasPitInformationIssue*	This dimension groups the boolean quality flags derived during the Data Cleaning phase. It is used to filter laps according to the type of information that is missing or unreliable. For example, laps with *HasSpeedInformationIssue* equal to true should not be used for speed-based analysis.
Lap Type (DD)	*LapType*	This dimension is used to distinguish normal laps, out-laps, in-laps, and pit-affected laps.
Lap Number (DD)	*LapNumber*	This dimension represents the progressive lap number completed by a driver within a session.

The entries marked `DD` are degenerate dimensions stored directly in `fact_lap_performance`. They do not generate separate physical tables.

The *Lap Data Quality* dimension is introduced as a technical quality dimension. It does not describe the sporting context of the lap, but the reliability and completeness of the information available for that lap.

The attributes of this dimension are boolean flags derived during the Data Cleaning phase. They indicate whether a specific analytical area is affected by missing, invalid, or outlier information.

This dimension can be interpreted as a *junk dimension* in the Star Schema. A junk dimension groups together several low-cardinality technical flags that are useful for filtering and data selection, but that do not justify the creation of many separate dimensions.

Some dimensions, such as *Tyre Context* and *Lap Weather Context*, are created by grouping several related attributes into a single analytical dimension.

This choice is motivated by the fact that these attributes describe the same contextual aspect of the lap. *Tyre Context* describes the tyre condition, while *Lap Weather Context* describes the aligned environmental condition.

Therefore, they are not direct copies of single entities in the reconciled schema, but semantic dimensions created after pruning, grafting, and derived-attribute definition. This makes the dimensional model clearer and more useful for OLAP queries, because related attributes are analyzed together instead of being scattered across the fact table or split into many small dimensions.

Measure Definition

In the *Lap Performance* fact, the selected measures must quantify the performance of a single lap.

The main numerical attributes retained as measures are lap time, sector times, speed values, position, and some derived count-based indicators.

The following table summarizes the main measures selected for the *Lap Performance* fact.

Measure Attribute
`LapTime_ms`
`Sector1Time_ms`
`Sector2Time_ms`
`Sector3Time_ms`
`SpeedI1`
`SpeedI2`
`SpeedFL`
`SpeedST`
`Position`

All the selected measures are non-additive measures. This means that they cannot be meaningfully aggregated using the `SUM` operator across dimensions.

These measures are instead analyzed using aggregation functions such as `AVG`, `MIN`, and `MAX`.

In particular, `Position` is an ordinal measure. For this reason, even when `AVG`, `MIN`, or `MAX` are used, the result must be interpreted carefully.

Glossary of Measures

Analytical Indicator	**Formula**	**Meaning**
Average Lap Time	`AVG(lap_time_ms)`	Average lap time for the selected analytical group.
Best Lap Time	`MIN(lap_time_ms)`	Fastest lap time in the selected analytical group.
Worst Lap Time	`MAX(lap_time_ms)`	Slowest lap time in the selected analytical group.

Analytical Indicator	Formula	Meaning
Average Sector Time	AVG(sector1_time_ms), AVG(sector2_time_ms), AVG(sector3_time_ms)	Average sector time for Sector 1, Sector 2, or Sector 3.
Best Sector Time	MIN(sector1_time_ms), MIN(sector2_time_ms), MIN(sector3_time_ms)	Fastest sector time for Sector 1, Sector 2, or Sector 3.
Worst Sector Time	MAX(sector1_time_ms), MAX(sector2_time_ms), MAX(sector3_time_ms)	Slowest sector time for Sector 1, Sector 2, or Sector 3.
Average Speed	AVG(speed_i1), AVG(speed_i2), AVG(speed_fl), AVG(speed_st)	Average speed measured at the intermediate speed traps, finish line, or main speed trap.
Maximum Speed	MAX(speed_i1), MAX(speed_i2), MAX(speed_fl), MAX(speed_st)	Maximum speed measured at the selected speed point.
Average Position	AVG(position)	Average position associated with the selected laps. This indicator must be interpreted carefully because position is an ordinal value.
Best Position	MIN(position)	Best position reached in the selected analytical group. Lower values indicate better positions.
Worst Position	MAX(position)	Worst position reached in the selected analytical group. Higher values indicate worse positions.

Workflow Session Result Fact

Shared relations from the Lap Performance workflow

The *Session Result* fact shares several structural relations with the previously discussed *Lap Performance* fact. In particular, the relations *Driver*, *Team*, *Session*, *Grand Prix*, *Circuit*, and *Season* remain analytically relevant also for this second fact.

For this reason, the detailed description of their attributes, together with the corresponding pruning choices and motivations, is not repeated in this section. Those considerations have already been presented in the workflow of the *Lap Performance* fact and remain valid here as well.

Therefore, in the present section, only the relations and attributes that are specific to the *Session Result* fact, or that require additional discussion in this second analytical context, are reported.

Session Result Table

A relevant characteristic of this relation is that it contains both qualifying results and race results. For this reason, many attributes are optional depending on the type of session. Some attributes are meaningful only for qualifying sessions, while others are meaningful only for race sessions.

We indicate in this table only the descriptive attributes.

Attribute	Meaning
Position	Final position obtained by the driver in the session.

Attribute	Meaning
ClassifiedPosition	Official classification of the driver in the session result. It may contain an integer value for classified drivers or categorical codes in race contexts. In the current reconciled extract, it is mainly informative for Race sessions and may be empty in Qualifying rows.
GridPosition	Starting position of the driver on the grid. Only for Race sessions.
Q1_ms	Best time obtained by the driver in Q1, expressed in milliseconds. Only for Qualifying sessions.
Q2_ms	Best time obtained by the driver in Q2, expressed in milliseconds. Only for Qualifying sessions.
Q3_ms	Best time obtained by the driver in Q3, expressed in milliseconds. Only for Qualifying sessions.
Time_ms	Time-related session result only for Race sessions.
Status	Textual status describing how the driver finished the race. Only for Race sessions.
Points	Number of championship points assigned to the driver for the Race sessions.
Laps	Number of laps completed by the driver. Only in Race sessions.

Note on `Time_ms`, `Status`, and `ClassifiedPosition` These three attributes require special attention because they are less straightforward than the other session result attributes.

ClassifiedPosition is a mixed attribute. In normal cases, it contains the official final classification of the driver as an integer value. However, it may also contain categorical codes such as `R` (retired), `D` (disqualified), `E` (excluded), `W` (withdrawn), `F` (failed to qualify), or `N` (not classified). For this reason, it is not a purely numeric attribute.

Status is an open textual attribute. It describes how the driver finished the session or why the driver did not finish it. Possible values include examples such as `Finished`, `+ 1 Lap`, `Crash`, and `Gearbox`, but the set of possible values is not closed or fully standardized.

Time_ms is a race-related time attribute. In practice, in our dataset, it does not behave as a fully homogeneous measure for all drivers. In some cases, it appears as the total race time of the winner, while in other cases it appears as a gap relative to the leader. Therefore, the attribute is useful, but its current name and raw meaning are too ambiguous for direct use in the conceptual model.

For this reason, the adopted solution is the following: the raw attributes are pruned. `ClassifiedPosition` and `Status` can later be transformed into cleaner derived categorical variables, while `Time_ms` can be replaced by a more explicit derived analytical measure, for example a gap-based measure.

The presence of many null values in the `result` relation is mainly due to the fact that the same relation stores two different kinds of sessions: Qualifying and Race.

Therefore, these null values must not automatically be interpreted as missing or dirty data. In many cases, they are semantically correct null values.

This characteristic of the `result` relation is important for the later multidimensional design.

The *Session Result* fact contains attributes whose meaning depends on the session type. This does not make the fact invalid, but it requires careful modeling.

In particular:

- qualifying-related attributes should be interpreted only when `session_type = Q`;
- race-related attributes should be interpreted only when `session_type = R`;

Pruning Operations

Attribute	Motivation
ClassifiedPosition	Removed because it is a mixed attribute that combines numeric and categorical values.
Status	Removed because it is an open textual attribute and not a clean categorical variable.
Time_ms	Removed because, in its raw form, it is not a fully homogeneous measure across all rows.

Grafting Operation

In the *Session Result* workflow, the root of the attribute tree is `Result_Id`.

In this second attribute tree, the *Lap Performance* fact is not considered. Therefore, `Lap_Id` does not appear in the *Session Result* attribute tree. The structure starts from the result-level fact and follows the relationships that are coherent with the grain of the result fact.

Before grafting, the result fact was connected to the session branch through `Session_Id`. The `Session_Id` node connected the result-level fact to the higher-level event hierarchy represented by `GrandPrix_Id`.

The initial structure was:

`Result_Id` → `Session_Id` → `GrandPrix_Id`

After pruning, the only analytically relevant attribute retained from the session node was `SessionType`. Since `Session_Id` is a technical identifier and not a useful aggregation level for the *Session Result* fact, it was removed through grafting.

After grafting, `SessionType` becomes a direct attribute of `Result_Id`, while `GrandPrix_Id` becomes a direct child of `Result_Id`.

The resulting structure is:

`Result_Id` → `SessionType`

and

`Result_Id` → `GrandPrix_Id`.

This operation preserves the information needed to distinguish the type of session and to reach the Grand Prix, Circuit, and Season hierarchy, while removing an intermediate technical node from the conceptual model.

Derived Attributes

Some attributes of the *Result* relation are analytically useful, but their raw form is not suitable to be used directly in the conceptual model.

This is the case of `classified_position`, `time_ms`, and `status`. These attributes require a transformation because they may be ambiguous, mixed, redundant, or too textual in their original form.

The goal of this step is not to discard useful information, but to transform raw result attributes into cleaner analytical attributes that are coherent with the grain of the *Session Result* fact.

Therefore, the derived attributes described below are attached directly to `Result_Id`, because they describe the final result obtained by a driver in a specific session.

Derived Attribute	Source Attribute	Analytical Meaning
ResultClassificationCategory	result.classified_position	Categorical attribute, it avoids duplicating the numeric position and describes only the classification state of the driver.
GapToLeader_ms	`result.time_ms`	Gap from the race leader expressed in milliseconds. The winner has value 0, while comparable classified drivers use the original gap value. Retired or non-comparable drivers may have a null value.
ResultStatusCategory	`result.status`	Categorical attribute derived from the raw textual status. It groups several textual values into stable analytical categories.

The raw attribute `classified_position` is pruned in its original form because it is a mixed attribute.

In Race sessions, it may contain numeric values, but it may also contain categorical codes such as retired, disqualified, excluded, withdrawn, or not classified. Moreover, when it contains numeric values, it repeats the same information already represented by *ResultPosition*.

For this reason, `classified_position` is not used to create another numeric position attribute. Instead, it is used to derive *ResultClassificationCategory*.

Possible values are:

- *Classified*
- *Retired*
- *Disqualified*
- *Excluded*
- *Withdrawn*
- *NotClassified*
- *FailedToQualify*
- *NotApplicable*

The value *Classified* is used when the original `classified_position` contains a regular numeric classification. The numeric value itself is not repeated, because it is already represented by *Position*.

The value *NotApplicable* is used for Qualifying sessions, where `classified_position` is normally not defined.

The raw attribute `time_ms` is pruned from the formal attribute tree because its meaning is not homogeneous across all rows.

In Race sessions, the winner usually has the total race time, while the other classified drivers may have the gap from the winner. In Qualifying sessions, this attribute is normally null.

For this reason, `time_ms` is replaced by a clearer derived measure called *GapToLeader_ms*.

The transformation rule is:

- for the race winner, *GapToLeader_ms* is set to 0;

- for classified drivers with a comparable race gap, the original `time_ms` value is interpreted as the gap from the winner;
- for retired, lapped, or non-comparable drivers, the value may be set to null;
- for Qualifying sessions, the value is also null.

This derived measure is more coherent because it always represents the same concept: the temporal gap from the race leader.

The raw attribute `status` is pruned from the formal attribute tree because it is an open textual field and its values are not fully standardized.

For example, it may contain values such as `Finished`, `+1 Lap`, `+2 Laps`, `Retired`, `Accident`, `Collision`, `Gearbox`, `Brakes`, or `Electrical`.

Keeping this raw attribute in the fact schema would introduce too many sparse textual values. For this reason, the information contained in `status` is transformed into a categorical attribute called *ResultStatusCategory*.

Possible values are:

- *Finished*
- *Lapped*
- *Accident*
- *Collision*
- *PowerUnitIssue*
- *MechanicalIssue*
- *Retired*
- *Other*
- *NotApplicable*

The category *PowerUnitIssue* is used for problems related to the engine or power unit. The category *MechanicalIssue* is used for other technical problems, such as gearbox, brakes, hydraulics, or similar failures.

The category *Lapped* is used for values such as `+1 Lap` or `+2 Laps`. This keeps the information that the driver finished behind the leader without introducing a separate *LapsBehind* measure.

Rare or unclear values can be grouped under *Other*. The value *NotApplicable* is used for Qualifying sessions, where the race status is normally not defined.

The *Weather* relation is not included as a direct branch of the formal attribute tree of the *Session Result* fact.

The reason is that weather data are time-dependent at session level. One session can be associated with multiple weather observations collected at different time instants. Therefore, the relationship:

`Session_Id` → `Weather`

does not determine one unique weather record.

However, weather information is still useful for the *Session Result* fact, because environmental conditions may influence the final outcome of a qualifying or race session.

For this reason, the ETL process aggregates weather observations at session level and derives attributes coherent with the grain of the *Session Result* fact.

Derived Attribute	Source Attribute	Analytical Meaning
RainFlag	`weather.rainfall`	Boolean flag indicating whether rainfall occurred during the session. It is useful to distinguish dry and wet sessions.
AvgAirTemp	`weather.air_temp`	Average air temperature during the session. It provides general environmental context for the session result.
AvgTrackTemp	`weather.track_temp`	Average track temperature during the session. It is useful because track temperature can influence tyre behaviour and race or qualifying performance.
AvgWindSpeed	`weather.wind_speed`	Average wind speed during the session. It provides additional context about external conditions that may influence the session outcome.

These attributes are attached directly to `Result_Id` in the edited attribute tree. Although they are computed at session level, each result row belongs to exactly one session. Therefore, after the ETL process, they are coherent with the grain of the *Session Result* fact.

The *TrackStatus* relation is not included as a direct branch of the formal attribute tree of the *Session Result* fact.

The reason is that track-status data are time-dependent at session level. During the same session, the track can move through different states, such as all-clear, yellow flag, safety car, virtual safety car, or red flag. Therefore, a session does not determine a single track-status record.

Although this information may be useful in some detailed race analyses, it is not retained as a derived attribute in the current *Session Result* fact.

The reason is that the *Session Result* fact focuses on the final result obtained by a driver in a session, such as final position, qualifying times, grid position, points, laps completed, and gap to the leader.

Simple derived attributes such as *SafetyCarPresence*, *VSCPresence*, *RedFlagPresence*, or *YellowFlagCount* would only indicate whether a given track-status event occurred during the session. However, they would not explain how that event affected a specific driver's final result.

For example, knowing that a Virtual Safety Car occurred during a race does not indicate whether a driver gained or lost time, whether the event happened before or after a pit stop, or whether it had a relevant impact on the final classification.

For this reason, track-status-derived attributes are not included in the current *Session Result* fact schema. They are considered optional contextual information for possible future extensions, but they are not necessary for the current analytical scope.

Data-Cleaning-Derived Quality Attributes

After the Data Cleaning phase, some additional boolean attributes are introduced for the *Session Result* fact.

These attributes are not taken directly from the original *Result* relation. They are derived from the focused missing value analysis and from some cleaning decisions applied to the `result` table.

Their purpose is not to measure performance. Instead, they describe whether some analytical information associated with a result row is missing, incomplete, or unreliable.

For this reason, they are treated as quality descriptors of the result row.

The information area `CLASSIFICATION_INFORMATION` is handled differently. If classification information is missing when required, the corresponding result row is removed during cleaning, because it cannot support the grain of the *Session Result* fact.

Therefore, no boolean flag is created for classification information in the cleaned `result` table.

The retained data-cleaning-derived attributes are:

Derived Quality Attribute	**Analytical Meaning**
HasQualifyingInformationIssue	Indicates that qualifying-related information is missing, incomplete, or suspicious. The row should not be used for complete qualifying time analysis without filtering.
HasRaceContextInformationIssue	Indicates that some race-context information is missing, invalid, or unreliable. The row may still be useful for classification analysis, but not for all race-context analyses.

The following table summarizes the derived attributes introduced for the *Session Result* fact.

Derived Attribute	**Derived From**	**Role**
ResultClassificationCategory	`result.classified_position`	Classification category
GapToLeader_ms	`result.time_ms`	Derived measure
ResultStatusCategory	`result.status`	Status category
RainFlag	`weather.rainfall`	Filter / context attribute
AvgAirTemp	`weather.air_temp`	Context attribute
AvgTrackTemp	`weather.track_temp`	Context attribute
AvgWindSpeed	`weather.wind_speed`	Context attribute
HasQualifyingInformationIssue	Data Cleaning output on `result`	Quality flag / Result Quality descriptor
HasRaceContextInformationIssue	Data Cleaning output on `result`	Quality flag / Result Quality descriptor

Dimension Definition

After the pruning and grafting operations, and after the introduction of the derived attributes, the final edited attribute tree is used to define the dimensions of the *Session Result* fact.

The following table summarizes the main dimensions selected for the *Session Result* fact.

Dimension	**Attributes**	**Motivation**
Driver	`Driver_Id`, *FirstName*, *LastName*, *Abbreviation*, *Nationality*	This dimension is used to compare session results among drivers.
Team	`Team_Id`, *TeamName*, *Nationality*	This dimension is used to compare the competitive performance of teams.
Grand Prix	`GrandPrix_Id`, *EventName*, *RoundNumber*, *EventDate*, *EventFormat*, *SeasonYear*, *CircuitId*, *CircuitName*, *Country*, *Location*, *GlobalCircuitCategory*, *Sector1Category*, *Sector2Category*, *Sector3Category*	This dimension allows the analysis of session results by Grand Prix, season, calendar order, and weekend format.

Dimension	Attributes	Motivation
Circuit hierarchy level	Circuit_Id, *CircuitName*, *Country*, *Location*, *GlobalCircuitCategory*, *Sector1Category*, *Sector2Category*, *Sector3Category*	Conceptual DFM hierarchy stored physically inside dim_grand_prix; no separate circuit dimension is created.
Session Type (DD)	*SessionType*	This dimension distinguishes qualifying sessions from race sessions.
Result Outcome	*ResultStatusCategory*, *ResultClassificationCategory*	This dimension describes the final outcome of the session result.
Result Weather Context	*RainFlag*, *AvgAirTempClass*, *AvgTrackTempClass*, *AvgWindSpeedClass*	This dimension describes the average environmental context of the session using categorical attributes.
Result Data Quality	*HasQualifyingInformationIssue*, *HasRaceContextInformationIssue*	This technical junk dimension groups boolean quality flags derived from the Data Cleaning phase. It is used to filter result rows according to the reliability of qualifying and race-context information.

The entry marked DD is stored directly as session_type in fact_session_result; it does not generate a physical session-type dimension. The categorical attributes of Result Weather Context are stored in dim_result_weather_context and referenced through result_weather_context_key.

In addition to the main business dimensions, the model includes a technical *Result Data Quality* dimension.

This dimension is a junk dimension because it groups low-cardinality boolean quality flags derived from the Data Cleaning phase. These flags are not measures and they do not represent a real business entity. They are included to support OLAP filtering and to avoid using result rows in analyses for which some important information is missing or unreliable.

Measure Definition

The following table summarizes the main measures selected for the *Session Result* fact.

Measure Attribute
Position
GridPosition
Points
Laps
Q1_ms
Q2_ms
Q3_ms
GapToLeader_ms

The selected measures do not all have the same aggregation behavior.

Points is an additive measure. It can be meaningfully aggregated using SUM, for example to compute the total points obtained by a driver, by a team, or during a season.

Laps is also additive when the analytical goal is to count the total number of laps completed by a driver or a team over a set of sessions. However, it can also be analyzed using AVG, MIN, and MAX when the goal is to study race completion or reliability.

`Position` is an ordinal measure. For this reason, it cannot be meaningfully aggregated using the `SUM` operator. It can be analyzed using `MIN`, `MAX`, and, with caution, `AVG`. Lower values indicate better positions.

`Q1_ms`, `Q2_ms`, and `Q3_ms` are non-additive time measures. They are meaningful mainly for qualifying sessions and must not be summed. They are analyzed using `AVG`, `MIN`, and `MAX`.

`GapToLeader_ms` is a derived non-additive measure. It expresses the time gap between a driver and the session leader.

Glossary of Measures

The following glossary defines the analytical indicators that can be computed from the measures of the *Session Result* fact.

Analytical Indicator	**Formula**	**Meaning**
Total Points	`SUM(points)`	Total number of points obtained by the selected driver, team, Grand Prix, or season.
Average Points	`AVG(points)`	Average points obtained in the selected analytical group.
Total Completed Laps	`SUM(laps)`	Total number of laps completed in the selected analytical group. This indicator can support reliability-oriented analyses.
Average Completed Laps	`AVG(laps)`	Average number of laps completed by drivers or teams in the selected group.
Best Final Position	`MIN(position)`	Best final position reached in the selected analytical group. Lower values indicate better results.
Worst Final Position	`MAX(position)`	Worst final position reached in the selected analytical group. Higher values indicate worse results.
Average Final Position	`AVG(position)`	Average final position in the selected analytical group. This indicator must be interpreted carefully because position is an ordinal value.
Best Q1 Time	`MIN(q1_ms)`	Fastest Q1 time in the selected analytical group. This indicator is meaningful only for qualifying sessions.
Best Q2 Time	`MIN(q2_ms)`	Fastest Q2 time in the selected analytical group. This indicator is meaningful only for qualifying sessions.
Best Q3 Time	`MIN(q3_ms)`	Fastest Q3 time in the selected analytical group. This indicator is meaningful only for qualifying sessions.
Average Q1 Time	`AVG(q1_ms)`	Average Q1 time for the selected analytical group. Null values must be ignored because not all drivers or sessions have a valid Q1 time.

Analytical Indicator	**Formula**	**Meaning**
Average Q2 Time	`AVG(q2_ms)`	Average Q2 time for the selected analytical group. Null values must be ignored because only drivers reaching Q2 have a valid value.
Average Q3 Time	`AVG(q3_ms)`	Average Q3 time for the selected analytical group. Null values must be ignored because only drivers reaching Q3 have a valid value.
Average Gap to Leader	`AVG(gap_to_leader_ms)`	Average time gap from the leader in the selected analytical group. Lower values indicate better relative performance.
Minimum Gap to Leader	`MIN(gap_to_leader_ms)`	Smallest gap from the leader in the selected analytical group. This indicator is useful for identifying the closest performances.
Result Count	`COUNT(result_id)`	Number of session-result records in the selected analytical group.
Points Finish Count	`COUNT(result_id) WHERE points > 0`	Number of results in which the driver or team scored points.
Podium Count	`COUNT(result_id) WHERE position <= 3`	Number of podium finishes in the selected analytical group.
Win Count	`COUNT(result_id) WHERE position = 1`	Number of session wins in the selected analytical group.